



Project Presentation

DevOps CI/CD Implementation : dataDash

**Automated pipeline with integrated security scanning and
Kubernetes(k3s) deployment.**

**Presented By.
Aadarsha Khadka**

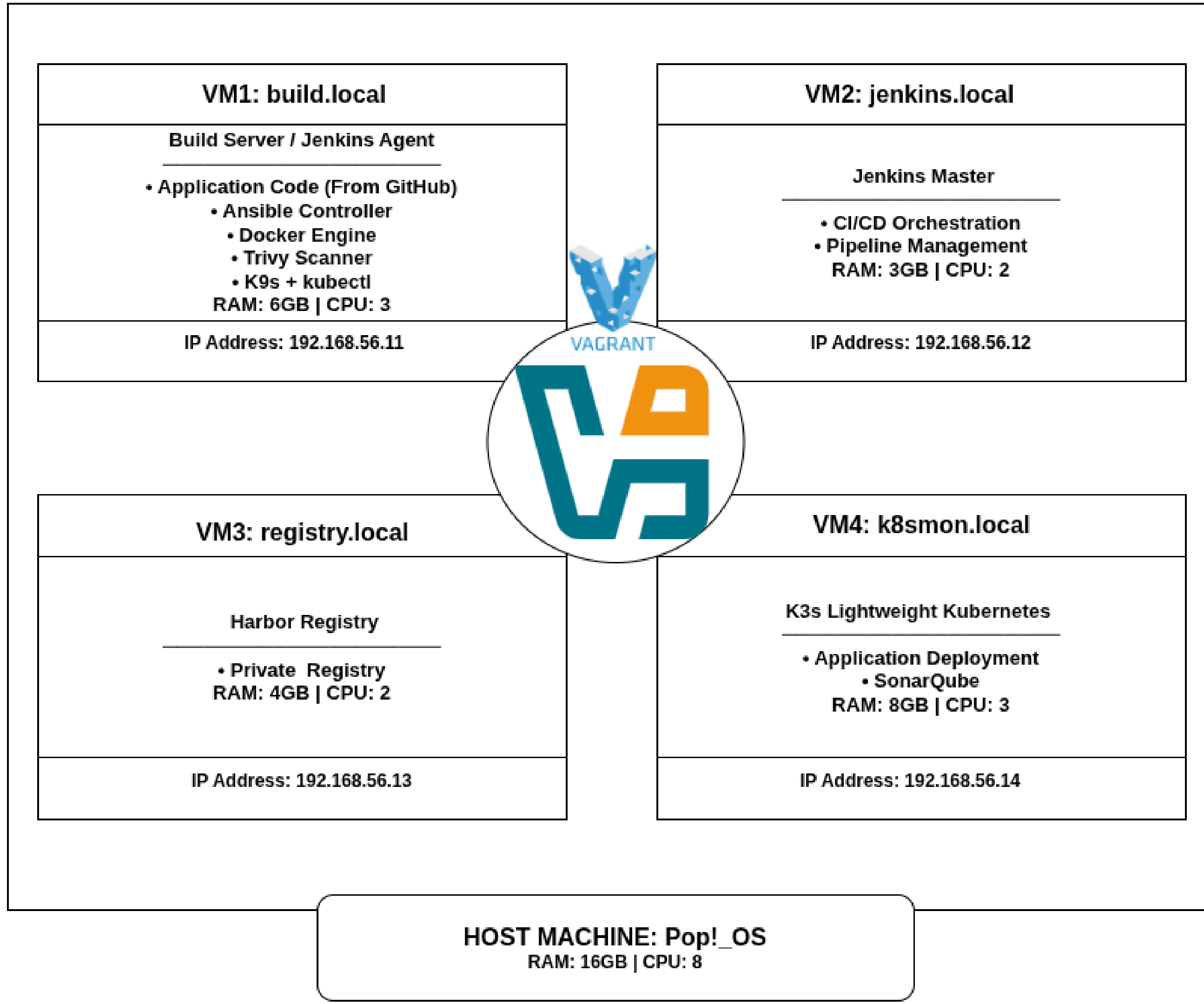
Objectives

- **Implement automated CI/CD pipeline**
- **Integrate static analysis and container scanning**
- **Deploy securely to K3s cluster**
- **Use private Harbor registry for image management**

Technology Stack

- **Jenkins for CI/CD**
- **SonarQube for static code analysis**
- **Trivy for container security**
- **Harbor as private registry**
- **K3s for Kubernetes deployment**
- **Docker for container builds**
- **Ansible for Configuring**
- **Git for VCS, GitHub for remote repo hosting**
- **K9s for cluster inspection**

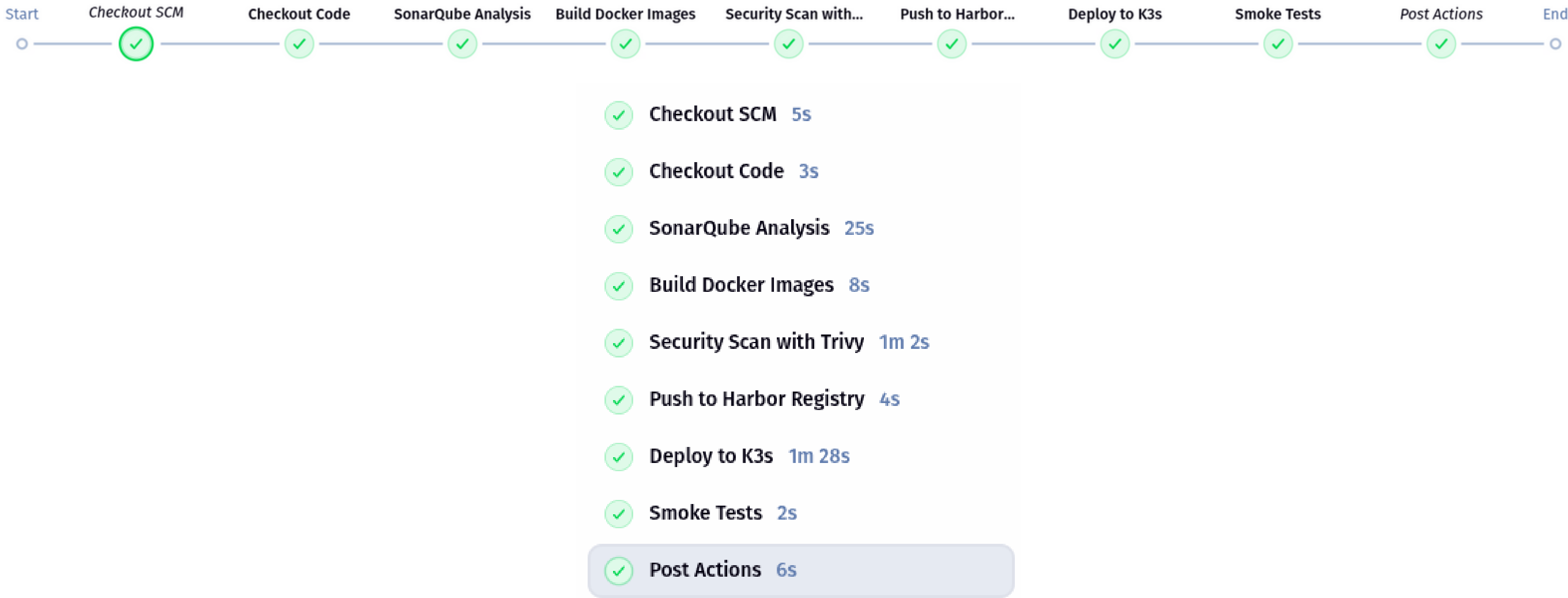
Architecture Overview



Jenkinsfile Structure

- **Automated stages: checkout, analysis, build, scan, push, deploy**
- **Creates namespace and secrets dynamically**
- **Performs rollout checks with debugging**
- **Runs smoke tests**
- **Sends HTML email notifications**
- **Cleans Docker images after pipeline**

CI/CD Pipeline Overview



Security Implementation

- **SonarQube static code analysis**
- **Trivy scan for HIGH/CRITICAL vulnerabilities**
- **Jenkins credential store for all secrets**
- **Secure Harbor registry authentication**

Kubernetes Deployment Model

- Backend and frontend as separate deployments
- Namespace and secrets auto-created
- Services, ingress, rolling updates handled
- Deployment validated with rollout status checks

Runtime Inspection

- **K9s used for pod logs and cluster inspection at runtime**
- **Monitoring stack excluded due to resource limitations**

Smoke Testing

- **Backend health check via curl inside pod**
- **Frontend readiness check using kubectl wait**
- **Validates deployment success before pipeline completes**

Notifications & Reporting

Build Successful!

Project: dataDash-pipeline
Build #: 33
Status: SUCCESS
Branch: origin/develop
Duration: 2 min 41 sec and counting
Build URL: <http://jenkins.local:8080/job/dataDash-pipeline/33/>

dataDash | Managed by the DevOps Team

Build Failed!

Project: dataDash-pipeline
Build #: 24
Status: FAILURE
Branch: origin/develop
Duration: 1 min 26 sec and counting
Build URL: <http://jenkins.local:8080/job/dataDash-pipeline/24/>

Please review the logs and fix the issue immediately.

[View Console Output](#)

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Operational Flow Summary And Live Demo

- 1. Code push → GitHub**
- 2. Manual trigger in Jenkins (Build Now)**
- 3. SonarQube: Static analysis**
- 4. Image build + Security Scan**
- 5. Push to Harbor**
- 6. Deploy to K3s**
- 7. Smoke tests**
- 8. Notifications**

Key Outcomes

- **Automated CI/CD pipeline**
- **Integrated DevSecOps steps**
- **Secure image management**
- **Reliable K3s deployment workflow**

What's next?

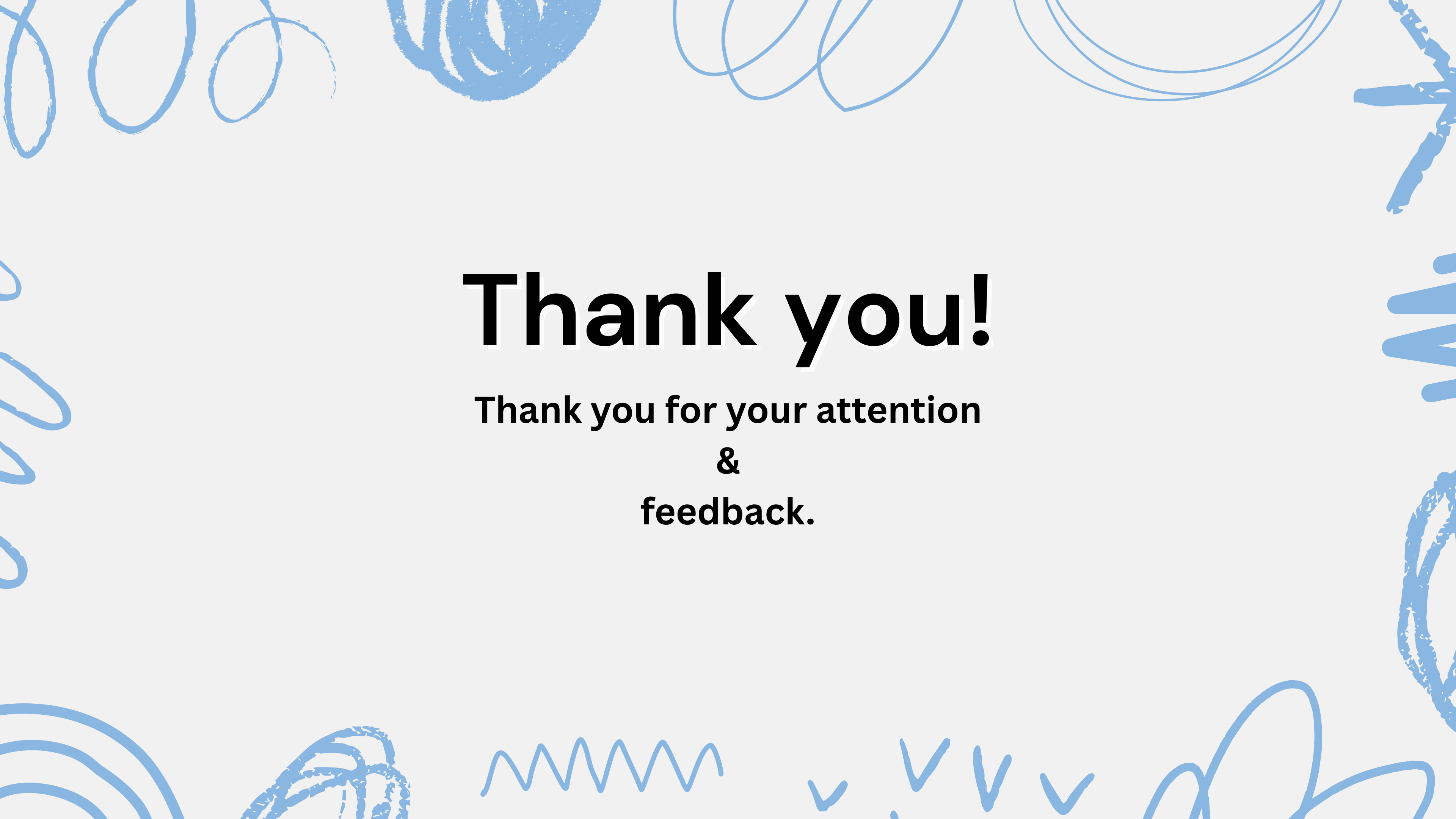
- **Add monitoring & observability: Prometheus/Grafana**
- **Implement centralized logging**
- **Enable auto-scaling**
- **Expand automated testing coverage**

Conclusion

- **A complete CI/CD pipeline with security integrated at every stage.**
- **Monitoring and advanced observability planned for future growth.**

Q & A

Questions?

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Thank you!

**Thank you for your attention
&
feedback.**